

Junkun Liu

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Education

Johns Hopkins University

Baltimore, MD

B.S. in Applied Mathematics & Statistics and Computer Science / GPA: 3.74/4.00

Expected May 2028

Relevant Coursework: Machine Learning & Deep Learning, NLP: Self-Supervised Models, Software System Design, Computer System Fundamentals, Data Structures & Algorithms, Probability, Optimization

Experience & Projects

LaunchStack – Tech Lead, Founder Operating System

Jun 2026 – Present

- Lead **architecture and engineering delivery** for the Founder Operating System, decomposing work into implementation plans and coordinating **code/PR review** across the team.
- Architected **document-change and Founder Weekly Review pipelines** that align artifact versions and synthesize **source-grounded evidence** from documents, GitHub activity, customer feedback, and team context.

StepFun – Algorithm Engineer Intern

Jun 2025 – Aug 2025

- Built and evaluated **LLM-based web agents** for multi-step browser tasks involving tool coordination, navigation, and **dynamic DOM changes**.
- Integrated structured agent workflows and **Azure OpenAI-compatible models** into evaluation pipelines; diagnosed **failure modes** including navigation loops, tool misuse, ambiguous elements, and inconsistent state transitions.

AfterShock – Project Lead, HopHacks 2026

Sep 2026

- Conceived the project and led a **4-person team** to a working demo in **36 hours**, winning **1st place among 27 teams** in the **Bloomberg Most Philanthropic Hack** track.
- Architected an **AI-assisted earthquake response system** that fuses conflicting multimodal reports into a shared operational state, coordinates search/rescue actions, and **dynamically reroutes teams** over real road networks.

Personal Action Agent – Independent AI Systems Project

Jun 2026 – Present

- Architected an **agent harness** that converts incoming and outgoing Gmail messages into persistent action records using structured outputs, **deterministic policy gates**, SQLite, and **append-only provenance**.
- Built a reliability and evaluation layer with **275 regression tests** and human-reviewed Gmail data; separated LLM interpretation from **deterministic control** of identity, lifecycle transitions, idempotency, and persistence.

Research

Genesis Mission – Undergraduate Researcher, Prof. Ziyang Li

Sep 2026 – Present

- Contribute to a **DOE-funded research project** on **verifiable computational physics code**, combining AI-based code generation with formal reasoning to improve scientific software reliability.
- Curate and analyze project datasets** and study **Lean** to support formal specification and validation workflows.

Dynamic Taint Analysis Research – Second-listed Author, Manuscript in Preparation

Jun 2026 – Jul 2026

- Coauthored research on **dynamic taint analysis across CPython, SpiderMonkey, and Chromium**; implemented propagation through calls, properties/indexes, concatenation, and native string operations.
- Built **reproducible full-application CVE validations** tracing tainted inputs to DOM and code-execution sinks; analyzed propagation DAGs, **source-to-sink coverage**, and runtime limitations.

Leadership & Teaching

Johns Hopkins University PILOT Program – Head PILOT Leader

Jul 2026 – Present

- Promoted after one year**; oversee the Honors Discrete Mathematics group and lead weekly **proof-based problem-solving sessions for 10–15 students**.

Johns Hopkins University – Teaching Assistant

Fall 2026 – Present

- Selected as a TA after earning an A+** in Mathematical Foundations of Computer Science; support instruction in proofs, discrete structures, and foundational CS reasoning.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL **AI/ML:** LLM Agents, RAG, Structured Outputs, Agent Evaluation, Embeddings, PyTorch, Scikit-learn

Systems/Research: SQLite, State Machines, Dynamic Taint Analysis, Browser Automation, CVE Validation, Docker

Web/Data: React, Next.js, Node.js, PostgreSQL, Git, AWS